



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	"RetailIQ" – Full-Cycle E-Commerce Customer & Sales Analytics	
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Abstract

The rapid growth of e-commerce generates large amounts of customer and sales data, making it difficult for businesses to understand purchasing patterns, customer behavior, and overall sales performance. The data may contain different customer, purchase, and shopping behavior attributes, making proper preprocessing and analysis necessary. Therefore, an effective system is needed to transform this data into meaningful insights.

Existing e-commerce solutions mainly focus on sales reports, customer analysis, or visualization of historical data. Some approaches also use techniques such as RFM analysis and customer segmentation to identify different types of customers. However, these activities are often handled separately. RetailIQ aims to provide an integrated Data Science approach that combines data preprocessing, sales analysis, customer behavior analysis, and machine learning-based customer segmentation in one system.

The proposed RetailIQ system uses an E-Commerce Customer Behavior and Purchase dataset and follows a complete Data Science workflow. The data will be cleaned and explored to identify customer, sales, and purchasing patterns. Relevant features will be engineered to represent customer purchasing behavior, and K-Means clustering will be used to group customers with similar characteristics. The results will be evaluated and presented through an interactive dashboard containing sales insights, product performance, customer behavior, and customer segments.

In the future, RetailIQ can be extended with advanced prediction and recommendation features. The system can be enhanced to predict repeat purchases, identify customers who may stop purchasing, forecast sales, and recommend suitable products. These extensions can make RetailIQ a more intelligent and useful decision-support system for e-commerce businesses.